

Converting Conic Sections to Standard Form

Algebra & Pre-Calculus Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the type of conic section from its general equation
- Use the completing the square technique to rewrite general equations in standard form
- Extract key features such as vertex, center, and orientation from the standard form

Problems

1. Identify the conic section represented by the equation below. Explain how you know.

$$y^2 - 8x + 4y + 4 = 0$$

2. Identify the conic section represented by the equation below.

$$x^2 + y^2 - 6x + 4y - 12 = 0$$

3. Complete the square for the expression below and write it as a perfect square trinomial in factored form.

$$y^2 + 10y$$

4. Complete the square for the expression below and write it in factored form.

$$x^2 - 12x$$

5. Convert the equation below to standard form. Then state the vertex and the direction the parabola opens.

$$y^2 + 6y - 4x + 33 = 0$$

6. Convert the equation below to standard form of a parabola. Then identify the vertex.



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$$x^2 - 4x - 8y + 28 = 0$$

7. Convert the equation below to standard form. Then identify the conic section and its center.

$$x^2 + y^2 - 6x + 4y - 12 = 0$$

8. Convert the equation below to standard form of an ellipse. Then identify the center.

$$x^2 + 4y^2 + 6x - 8y - 3 = 0$$

9. Convert the equation below to standard form of a hyperbola. Then identify the center.

$$9x^2 - 4y^2 - 18x + 24y - 63 = 0$$

10. Convert the equation below to standard form. Identify the conic section, its center or vertex, and describe its orientation.

$$4x^2 + y^2 - 16x + 6y - 39 = 0$$



Converting Conic Sections to Standard Form — Answer Key

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Answer Key

1. Answer: Parabola — only one variable is squared

- Check which variables have an exponent of 2
- Only y is squared, so this is a parabola

2. Answer: Circle — both variables are squared with equal coefficients

- Both x and y are squared with the same coefficient (1)
- Equal coefficients indicate a circle

3. Answer: $(y + 5)^2 - 25$

- Take half of 10: $10 \div 2 = 5$
- Square it: $5^2 = 25$
- Add and subtract 25: $y^2 + 10y + 25 - 25 = (y + 5)^2 - 25$

4. Answer: $(x - 6)^2 - 36$

- Take half of -12 : $-12 \div 2 = -6$
- Square it: $(-6)^2 = 36$
- Add and subtract 36: $x^2 - 12x + 36 - 36 = (x - 6)^2 - 36$

5. Answer: $(y + 3)^2 = 4(x - 6)$; vertex $(6, -3)$; opens right

- Rearrange: $y^2 + 6y = 4x - 33$
- Complete the square: $y^2 + 6y + 9 = 4x - 33 + 9$
- Factor: $(y + 3)^2 = 4x - 24$
- Factor right side: $(y + 3)^2 = 4(x - 6)$
- Vertex is $(6, -3)$; since y is squared and coefficient is positive, opens right

6. Answer: $(x - 2)^2 = 8(y - 3)$; vertex $(2, 3)$

- Rearrange: $x^2 - 4x = 8y - 28$
- Complete the square: $x^2 - 4x + 4 = 8y - 28 + 4$
- Factor: $(x - 2)^2 = 8y - 24$
- Factor right side: $(x - 2)^2 = 8(y - 3)$
- Vertex is $(2, 3)$; since x is squared, parabola opens upward

7. Answer: $(x - 3)^2 + (y + 2)^2 = 25$; circle with center $(3, -2)$

- Group: $(x^2 - 6x) + (y^2 + 4y) = 12$
- Complete the square for x : add 9; for y : add 4
- $(x^2 - 6x + 9) + (y^2 + 4y + 4) = 12 + 9 + 4$
- $(x - 3)^2 + (y + 2)^2 = 25$



- This is a circle with center $(3, -2)$ and radius 5

8. Answer: $\frac{(x+3)^2}{16} + \frac{(y-1)^2}{4} = 1$; center $(-3, 1)$

- Group: $(x^2 + 6x) + 4(y^2 - 2y) = 3$
- Complete the square for x: add 9; for y inside parentheses: add 1 (multiply by 4 outside)
- $(x^2 + 6x + 9) + 4(y^2 - 2y + 1) = 3 + 9 + 4$
- $(x + 3)^2 + 4(y - 1)^2 = 16$
- Divide by 16: $(x+3)^2/16 + (y-1)^2/4 = 1$; center $(-3, 1)$

9. Answer: $\frac{(x-1)^2}{4} - \frac{(y-3)^2}{9} = 1$; center $(1, 3)$

- Group: $9(x^2 - 2x) - 4(y^2 - 6y) = 63$
- Complete the square: $9(x^2 - 2x + 1) - 4(y^2 - 6y + 9) = 63 + 9 - 36$
- $9(x - 1)^2 - 4(y - 3)^2 = 36$
- Divide by 36: $(x-1)^2/4 - (y-3)^2/9 = 1$
- This is a hyperbola with center $(1, 3)$

10. Answer: $\frac{(x-2)^2}{16} + \frac{(y+3)^2}{64} = 1$; ellipse, center $(2, -3)$, major axis vertical

- Group: $4(x^2 - 4x) + (y^2 + 6y) = 39$
- Complete the square: $4(x^2 - 4x + 4) + (y^2 + 6y + 9) = 39 + 16 + 9$
- $4(x - 2)^2 + (y + 3)^2 = 64$
- Divide by 64: $(x-2)^2/16 + (y+3)^2/64 = 1$
- Ellipse with center $(2, -3)$; since $64 > 16$, major axis is vertical

